

Every design choice has a source

Eight papers and repositories, and the feature each one shaped in AEGIS Twin

MICRO TWIN

A SUMO-Based Digital Twin for Conventional & EV Networks

arXiv 2507.10280 · Eclipse SUMO 1.27

SUMO is a credible twin substrate when it is checked against reality, corridor by corridor.

→ Hyderabad micro-twin on real OSM roads, plus a **Wasserstein** sim-vs-observed score per corridor

LIVE STREAMING

sidewalklabs / sumo-web3d

github.com/sidewalklabs/sumo-web3d

TraCI → WebSocket frame-diffs → browser is enough to animate a whole city live.

→ msgpack diffs at 5–10 Hz, lerped client-side to **60 fps** with deck.gl TripsLayer

MACRO ENGINE

SupplyNetPy + SimPy real-time / dynamic-des

arXiv 2607.09745 · jaehyeon-kim/dynamic-des

(s,S)/(R,Q) policies and disruptions in SimPy, validated against AnyLogistix, EOQ and newsvendor; live parameter updates.

→ Custom SimPy twin on wall-clock time, with **engine verification** tests before any number is trusted

RESILIENCE THEORY

Simchi-Levi: Time-to-Survive vs Time-to-Recover

MIT · TTS / TTR / Risk Exposure Index

A node is exposed when it runs out before its supply can recover.

→ The demo headline "**TTS 3.2 d < TTR 5 d**", TTS-vs-TTR in every node drill-down, REI ranking

RIPPLE EFFECT

Ivanov et al., Digital Supply Chain Twins: Managing the Ripple Effect

Springer · + PLOS One critical-node cascades

Disruptions propagate through the network topology, not one node at a time.

→ **Cascade simulation** on the lane graph, 3D Network Graph, auto-failover of demand zones

GPS SPOOFING

GPS-IDS + decomposition Kalman filter under spoofing

arXiv 2405.08359 · Sci. Rep. s41598-025-32863-5

Physics models plus state estimators catch both crude teleports and subtle drift.

→ Trust layers 4 & 6: teleport checks and **Kalman χ^2 gating**; red ghost vs green estimate on the map

FEED ANOMALIES

Disruption Detection for a Cognitive Digital Supply Chain Twin

arXiv 2309.14557 · + SupplyTwin-Simulation

Anomaly detection on supply feeds; validation-first ingestion with deterministic seeds.

→ Trust layer 8 (**IsolationForest + MAD**), strict schemas, seeded everything, evidence-gated AI

EVALUATION

Digital Twin Counterfactual Framework

arXiv 2604.01325

A fidelity hierarchy: from matching observations to getting decisions right.

→ Fidelity Lab: shadow predictions, calibration, and **counterfactual decision value** across 50 disruptions